like piece of flesh on the end of a long 'fishing rod', projecting from the top of the head. When a small prey fish comes near, the angler will dance its worm-like bait in front of the little fish, and lure it down to the region of the angler's own concealed mouth. Suddenly it opens its jaws, and the little fish is sucked in and eaten. The angler is telling a lie, exploiting the little fish's tendency to approach wriggling worm-like objects. He is saying 'Here is a worm', and any little fish who 'believes' the lie is quickly eaten.

Some survival machines exploit the sexual desires of others. Bee orchids induce bees to copulate with their flowers, because of their strong resemblance to female bees. What the orchid has to gain from this deception is pollination, for a bee who is fooled by two orchids will incidentally carry pollen from one to the other. Fireflies (which are really beetles) attract their mates by flashing lights at them. Each species has its own particular dot-dash flashing pattern, which prevents confusion between species, and consequent harmful hybridization. Just as sailors look out for the flash patterns of particular lighthouses, so fireflies seek the coded flash patterns of their own species. Females of the genus *Photuris* have 'discovered' that they can lure males of the genus *Photinus* if they imitate the flashing code of a *Photinus* female. This they do, and when a *Photinus* male is fooled by the lie into approaching, he is summarily eaten by the *Photuris* female. Sirens and Lorelei spring to mind as analogies, but Cornishmen will prefer to think of the wreckers of the old days, who used lanterns to lure ships on to the rocks, and then plundered the cargoes that spilled out of the wrecks.

Whenever a system of communication evolves, there is always the danger that some will exploit the system for their own ends. Brought up as we have been on the 'good of the species' view of evolution, we naturally think first of liars and deceivers as belonging to different species: predators, prey, parasites, and so on. However, we must expect lies and deceit, and selfish exploitation of communication to arise whenever the interests of the genes of different individuals diverge. This will include individuals of the same species. As we shall see, we must even expect that children will deceive their parents, that husbands will cheat on wives, and that brother will lie to brother.

Even the belief that animal communication signals originally evolve to foster mutual benefit, and then afterwards become exploited by malevolent parties, is too simple. It may well be that all animal